

ON THE PROBABILITY A WEIGHTED BERNOULLI SUM EXCEEDS ITS MEAN

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ABSTRACT. Let $w_1, \dots, w_m$ be positive real weights summing to 1, and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables. If we let $X = \sum_{i=1}^m w_i v_i$, then we ask for which $p \in (0, 1)$ we have that

$$\mathbb{P}[X \geq \mathbb{E}[X]] \geq p?$$

In this short note, we observe a connection of this question with a version of the Manickam–Miklós–Singhi conjecture, which allows one to resolve it for sufficiently small values of p .

1. INTRODUCTION

Let $w \in \mathbb{R}_{>0}^m$ be a vector of positive weights satisfying $\sum_{i=1}^m w_i = 1$, and let $v \in \{0, 1\}^m$ be a random binary vector, in which every coordinate equals 1 independently and with some fixed probability p . Then, the random variable $X = \langle w, v \rangle$ takes values in $[0, 1]$ and satisfies $\mathbb{E}[X] = p$.

In this note, we study lower bounds on the probability that X exceeds p , i.e. its mean. In general, one cannot expect a lower bound better than p , because of the following example. If $m = 1$ and $w_1 = 1$, then $X = v_1$ and $\mathbb{P}[X \geq \mathbb{E}[X]] = \mathbb{P}[v_1 = 1] = p$. The following question asks for which value of p this example is indeed the worst one.

Question 1. *For which values of $p \in (0, 1)$ does the following hold? For every $w \in \mathbb{R}_{>0}^m$ such that $\sum_{i=1}^m w_i = 1$, if $v \in \{0, 1\}^m$ is a random vector with i.i.d. Bernoulli(p) coordinates, then*

$$\mathbb{P}[\langle v, w \rangle \geq p] \geq p.$$

To the best of our knowledge, the first result proved in this direction comes from the work of Alon, Emek, Feldman and Tennenholtz [?] on information leakage in adversarial games, where they have shown that $\mathbb{P}[X \geq \mathbb{E}[X]] \geq p(1-p)/10$ for all p , which is tight up to a constant factor for $p \leq 1/2$. One natural guess would be that Question ?? holds for all $p \leq 1/3$ and for $p = 1/2$. In Proposition ?? we show that the inequality does not hold for other values of p .

It is also worth noting that [?] already proposes a very elegant coupling argument showing that the answer to Question ?? is yes if p is of the form $p = \frac{1}{n}$ for some positive integer n . In fact, the proof in this note is inspired by their argument.

Let us also note that in the special case where all weights are equal (so that $mX \sim \text{Bin}(m, p)$), related lower bounds for $\mathbb{P}[\text{Bin}(m, p) \geq mp]$ have been studied; see, e.g., Greenberg–Mohri [?], Pelekis–Ramon [?], and Doerr [?]. Finally, let us mention that Question ?? is also vaguely reminiscent of Feige’s conjecture, which states that for independent positive random variables $X_1, \dots, X_n$, each of which has expectation 1, we have $\mathbb{P}[X_1 + \dots + X_n \geq n + 1] \leq 1 - 1/e$.

The purpose of this note is to explain an unexpected (at least to us) connection of this problem to the classical Manickam–Miklós–Singhi (MMS) conjecture on the number of nonnegative k -sums.

Conjecture 2. *Let n, k be positive integers, with $n \geq 4k$. For any real numbers $x_1, \dots, x_n$ whose sum is 0, at least $\binom{n-1}{k-1}$ of the k -element subsets $S \subseteq [n]$ satisfy*

$$\sum_{i \in S} x_i \geq 0.$$

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This question was first raised by Bier and Manickam [?], who proved Conjecture ?? when k divides n or when n is much larger than k , and this exact form of the conjecture was later put forward by Manickam and Miklós [?] and also by Manickam and Singhi [?]. In 2012, Tyomkyn [?] significantly improved the quantitative bounds, proving the conjecture for $n > k(4e \log k)^k$. Later, Alon, Huang and Sudakov [?] proved the polynomial bound and showed the conjecture when $n \geq 33k^2$. Finally, a linear bound $n \geq 10^{46}k$ was obtained by Pokrovskiy [?].

The main purpose of this note is to prove the following theorem. This theorem, combined with the result of Pokrovskiy [?], shows that the answer to Question ?? is positive for all $0 \leq p < 10^{-46}$. Moreover, it shows that conditionally on the Manickam-Miklós-Singhi conjecture being true, Question ?? holds for all $0 \leq p \leq 1/4$.

Theorem 3. *If there is a constant c such that Conjecture ?? holds for all $n \geq ck$, then Question ?? holds for all $p \in [0, 1/c)$.*

Interestingly, the same theorem would still hold true if one weakens Conjecture ?? to only ask about sequences $x_1, \dots, x_n$ where all negative entries have the same value.

2. PROOFS

Proof of Theorem ??. We split this proof into two parts - first, we will show the positive answer Question ?? for all rational numbers $p \in [0, 1/c)$, and then we will extend it to all $p \in [0, 1/c)$ via a simple limiting argument.

Thus, let us assume that we have $p = \frac{k_0}{n_0}$, a rational number, and let $w \in \mathbb{R}_{>0}^m$ be the corresponding weight vector. Let t be a large integer, and let us set $k = k_0 t$, $n = n_0 t$.

The crucial object of the proof is the random 0/1-matrix M , with n rows and m columns. To generate M , we will fill out every column with $n - 1$ zeros and exactly 1 one, with the position of the 1 being uniformly picked among the n possible rows, independently of all other decisions. If we denote the rows of M by $r_1, \dots, r_n$, we calculate the inner products $y_i = \langle w, r_i \rangle$, and we choose a k -element set $S \subset [n]$ uniformly at random. Finally, we set $Y = \sum_{i \in S} y_i$.

Claim 4. *The random variable Y has the same distribution as the inner product $X = \langle w, v \rangle$.*

Proof. Since $Y = \langle w, \sum_{i \in S} r_i \rangle$, it suffices to show that $\sum_{i \in S} r_i$ follows the same distribution as v . For any fixed set S and $j \in [m]$, the j -th coordinate of $\sum_{i \in S} r_i$ is 1 precisely if one of the rows corresponding to S contains a 1 in the j -th column, which happens exactly with probability $\frac{|S|}{n} = \frac{k}{n} = p$. Since the distribution of $\sum_{i \in S} r_i$ is the same as the distribution of v for any fixed set S , the same is true if S is chosen uniformly at random. $\square$

Claim 5. *We have*

$$\mathbb{P}[Y \geq p] \geq p.$$

Proof. Let us define $z_i = y_i - \frac{1}{n}$. First, observe that y_i either takes value 0 (if all coordinates of r_i are 0), or it takes value at least $\min_{j \in [m]} \{w_j\} > 0$. Further, observe that if t is sufficiently large, then $\frac{1}{n} = \frac{1}{n_0 t} < \min_{j \in [m]} \{w_j\}$. Therefore, the only distinct negative value which appears among z_i 's is $-\frac{1}{n}$.

Also, since the sum of coordinates of w is 1 and M contains exactly one 1 per column, we have

$$\sum_{i=1}^n y_i = \langle w, \sum_{i=1}^n r_i \rangle = \langle w, \mathbf{1} \rangle = 1.$$

Thus, $\sum_{i=1}^n z_i = 0$.

Finally, since $p = \frac{k}{n} < 1/c$, we have $n \geq ck$. Thus, if Conjecture ?? holds for $n \geq ck$, we can apply it to the numbers $z_1, \dots, z_n$ to conclude that a uniform random set $S \subset [n]$ of size k has nonnegative sum with probability at least $\frac{\binom{n-1}{k-1}}{\binom{n}{k}} = \frac{k}{n} = p$.

Since $\sum_{i \in S} z_i \geq 0$ is equivalent to $Y = \sum_{i \in S} y_i \geq \frac{|S|}{n} = p$, we conclude that with probability at least p over the choice of S we have $Y \geq p$, as we wanted to show. $\square$

Combining the two claims above shows that $\mathbb{P}[\langle v, w \rangle \geq p] = \mathbb{P}[Y \geq p] \geq p$ whenever p is a rational and at most $\frac{1}{c}$. To show the conjecture when p is irrational, let p_i be a sequence of rationals approaching p from below and observe that $\mathbb{P}[\langle w, v \rangle \geq p_i] \geq p_i$ (where the entries of v are still distributed according to $\text{Bernoulli}(p)$, as before). This inequality follows directly from our argument above, since v can be coupled with a vector v_i whose coordinates are distributed according to $\text{Bernoulli}(p_i)$, so that we always have that v dominates v_i on every coordinate. The conclusion is that

$$\mathbb{P}[\langle w, v \rangle \geq p] = \lim_{p_i \rightarrow p} \mathbb{P}[\langle w, v \rangle \geq p_i] \geq \lim_{p_i \rightarrow p} p_i = p \quad \square$$

Proposition 6. *The answer to Question ?? is negative for any $p \in (1/3, 1)$, with the exception of $p = 1/2$.*

Proof. Let us first consider the case $1/3 < p < 1/2$. In this case, choosing $w = (1/3, 1/3, 1/3)$, we see that $\langle w, v \rangle \geq p$ if and only if v has ones on at least 2 coordinates. Thus,

$$\mathbb{P}[\langle w, v \rangle \geq p] = 3p^2(1 - p) + p^3 = p(3p - 2p^2).$$

It is not hard to verify that this quantity is smaller than p for all $p \in (1/3, 1/2)$, since

$$p - p(3p - 2p^2) = p(1 - 3p + 2p^2) = p(1 - 2p)(1 - p) > 0.$$

If $1/2 < p < 1$, we have a similar example, with $w = (1/2, 1/2)$. Again, we see that $\langle w, v \rangle \geq p$ if and only if v has ones on at least 2 coordinates. Thus,

$$\mathbb{P}[\langle w, v \rangle \geq p] = p^2 < p. \quad \square$$

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REFERENCES

- [1] T. Bier and N. Manickam, *The first distribution invariant of the Johnson-scheme*, *SEAMS Bull. Math.* **11** (1987), 61–68.
- [2] N. Manickam and D. Miklós, *On the number of nonnegative partial sums of a nonnegative sum*, In *Colloq. Math. Soc. Janos Bolyai* **52** (1987), 385–392.
- [3] N. Manickam and N. M. Singhi, *First distribution invariants and EKR theorems*, *J. Combin. Theory Ser. A* **48** (1988), 91–103.
- [4] N. Alon, Y. Emek, M. Feldman, and M. Tennenholtz, *Adversarial leakage in games*, *SIAM J. Discrete Math.* **27** (2013), 363–385.
- [5] N. Alon, H. Huang, and B. Sudakov, *Nonnegative k -sums, fractional covers, and probability of small deviations*, *J. Combin. Theory Ser. B*, **102** (2012), 784–796.
- [6] M. Tyomkyn, *An improved bound for the Manickam–Miklós–Singhi conjecture*, *European J. Combin.* **33** (2012), 27–32.
- [7] S. Greenberg and M. Mohri, *Tight lower bound on the probability of a binomial exceeding its expectation*, *Statistics & Probability Letters* **86** (2014), 91–98.
- [8] C. Pelekis and J. Ramon, *A lower bound on the probability that a binomial random variable is exceeding its mean*, *Statistics & Probability Letters* **119** (2016), 305–309.
- [9] B. Doerr, *An elementary analysis of the probability that a binomial random variable exceeds its expectation*, *Statistics & Probability Letters* **139** (2018), 67–74.
- [10] A. Pokrovskiy, *A linear bound on the Manickam–Miklós–Singhi conjecture*, *J. Combin. Theory Ser. A* **133** (2015), 280–306.